

Multiple Choice (10 marks)

1. How did scientists arrive at the theory of plate tectonics?
 - (a) by determining the age of lava rising at ocean ridges
 - (b) by determining the composition of limestone deposits
 - (c) by determining the rate of weathering of mountain ranges
 - (d) by determining the type of sedimentary rock formed on land
2. A caterpillar eats an oak leaf. Which of the following best describes the energy transfer in this situation?
 - (a) Both the caterpillar and the leaf gain energy.
 - (b) Energy is transferred from the leaf to the caterpillar.
 - (c) Decomposers in the leaf obtain energy from the caterpillar.
 - (d) The oak tree gains energy when the caterpillar eats the leaf.
3. Which of the following must get nutrients from a transport system?
 - (a) amoeba
 - (b) euglena
 - (c) skin cell
 - (d) bacterial cell
4. What is the proper procedure for safely disposing of an acid solution after finishing an experiment?
 - (a) pour the solution down the sink
 - (b) neutralize the solution with a base
 - (c) dilute the solution with more acid
 - (d) absorb the liquid with paper towels
5. What is formed when a neutral atom gains an electron?
 - (a) A mixture
 - (b) An ion
 - (c) A molecule
 - (d) A metal

True / False (10 marks)

Answer True or False

6. True or False: Water vapor exists in the atmosphere as a visible liquid that can be observed floating in the air.
7. True or False: The ice cube melted due to heat transferred from the air and the pavement by conduction.
8. True or False: Comparing the skeletons of a fish that lived recently and one that lived a long time ago can illustrate evolutionary changes.
9. True or False: The albumen is the clear, thick fluid inside a poultry egg that provides protection for the developing embryo.
10. True or False: Desert animals are active at night to find more food illuminated by the moonlight.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the similarities and differences in the reproductive processes of euglena and paramecium, highlighting the key mechanisms involved in each organism.
12. Discuss the properties of a saltwater mixture used for cleaning contact lenses and explain why it is effective for this purpose.

End of Paper